Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-K Infectious Vascular Diseases

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-18-11-C01-BD1

Brain dump 1 · foundational retrieval: Build a vegetation from a normal valve

A competent response contains

· Normal endothelium resists bacterial attachment, so most cases need two things at once: a surface bacteria can grip and bacteria in the blood.

· Step 1: a turbulent high-velocity jet (regurgitant or stenotic), a ventricular septal defect, a prosthetic ring, or repeated injection of particulate material injures valve endothelium and exposes subendothelial collagen and tissue factor.

· Step 2 and 3: platelets and fibrin deposit on the injured area and form a small sterile mass, which is nonbacterial thrombotic endocarditis.

· Step 4: organisms entering during a transient bacteremia adhere to the deposit. Viridans streptococci make extracellular dextran (a glucan) and other adhesins that stick to fibrin and platelets, so they need a damaged valve.

· Staphylococcus aureus carries surface proteins that bind fibrinogen and fibronectin and can attach to and invade intact endothelium.

· Step 5 and 6: more fibrin covers the bacteria and shields them, and the growing vegetation sheds organisms into the blood continuously.

· The vegetation is avascular, so neutrophils penetrate it poorly, bacteria reach very high densities, and bacteremia is persistent rather than intermittent, which is why blood cultures stay positive.

· Staphylococcus aureus can skip the first three steps and infect a valve that was never damaged, which is why it produces acute disease on normal valves.

Distinctions that matter

· Adhesion to a sterile platelet and fibrin deposit is what viridans streptococci need; invasion of intact endothelium is what S. aureus can add.

· A vegetation is infected; nonbacterial thrombotic endocarditis is the sterile deposit that precedes it.

Common misconceptions

· Writing that bacteria settle on any valve they reach. Normal endothelium resists attachment, which is why turbulence and a sterile deposit usually come first.

· Explaining persistent bacteremia by saying the immune system is weak. The vegetation is avascular and fibrin covered, so neutrophils cannot get in.

Where it is taught: Section C01, first three paragraphs and the flow diagram of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C01-BD2

Brain dump 2 · integration and application: Reason from the setting to the organism

A competent response contains

· Patient 1: viridans streptococci (for example S. sanguinis, S. mutans) from mouth flora released at extraction, adhering to the deposit on a regurgitant mitral valve through dextran.

· Patient 2: Staphylococcus aureus from skin flora injected into the blood, reaching the right heart first, so the tricuspid valve is involved; Pseudomonas and Candida also occur in injection drug use.

· Patient 3: Staphylococcus epidermidis or another coagulase-negative staphylococcus, introduced at operation and growing as a biofilm on foreign material.

· Patient 4: enterococci, mainly Enterococcus faecalis, from urinary and gut flora entering the blood during manipulation.

· Patient 5: Streptococcus gallolyticus (formerly S. bovis), which is associated with colorectal cancer, so the colon is examined.

· Patient 6: culture-negative endocarditis from fastidious or intracellular organisms, the HACEK group, Coxiella burnetii, or Bartonella, which grow poorly in routine culture.

· Acute course: patients 2 and 3, from virulent organisms that destroy a valve over days. Subacute course: patients 1, 4, 5, and 6, smoldering for weeks with fever, weight loss, and fatigue.

· Virulence of the organism decides the course, not the valve, which is why the time line in a question points at the organism.

· Late prosthetic valve infection behaves more like native valve disease, with streptococci and S. aureus becoming common.

Distinctions that matter

· Early and late prosthetic valve infection have different organisms: early is skin flora from the operation, late resembles native valve disease.

· Negative cultures are a clue to the organism rather than evidence against endocarditis.

Common misconceptions

· Naming S. aureus for every case. The source of the bacteremia and the surface available decide the organism.

· Assuming injection drug use always means a left-sided valve. Injected organisms reach the right heart first and the tricuspid valve is the usual site.

Where it is taught: Section C01, organism table and the worked example of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C01-RT01

Retrieval exercise: Rebuild the vegetation sequence and mark where vegetations sit

A competent response contains

· Step 1: a turbulent jet or foreign material injures valve endothelium.

· Step 2: collagen and tissue factor are exposed and platelets and fibrin deposit.

· Step 3: a sterile deposit forms, which is nonbacterial thrombotic endocarditis.

· Step 4: during a transient bacteremia organisms adhere through dextran or fibrinogen and fibronectin binding proteins.

· Step 5: more fibrin covers the bacteria and shields them from neutrophils.

· Step 6: the vegetation grows and sheds organisms into the blood continuously.

· Vegetations sit along the line of valve closure: on the atrial surface of the mitral and tricuspid valves and the ventricular surface of the aortic and pulmonary valves.

· Those are the low-pressure sides, the surfaces that receive the regurgitant jet, which is why the deposit forms there.

· The mitral valve and then the aortic valve are involved most often because they work at higher pressures.

· In injection drug use the tricuspid valve is the most common site, because injected bacteria and particulate debris reach the right heart before anything else.

Distinctions that matter

· The vegetation forms on the downstream, lower pressure side of the valve, not on the side the pressure is generated.

· Higher working pressure explains left-sided predominance; route of entry explains the tricuspid exception.

Common misconceptions

· Placing mitral vegetations on the ventricular surface. The mitral regurgitant jet strikes the atrial surface, so the deposit forms there.

Where it is taught: Section C01, third paragraph and the flow diagram of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C01-Q01

Board-style question: Organism in right-sided endocarditis

Correct answer: C. Staphylococcus aureus

Why it is right: Skin flora are injected directly into the venous blood, Staphylococcus aureus surface proteins bind fibrinogen and fibronectin and can invade intact endothelium, so it infects the normal tricuspid valve and destroys it over days, and fragments of the vegetation lodge in the pulmonary arteries as cavitating septic emboli.

A is wrong: Streptococcus gallolyticus bacteremia is associated with colorectal cancer and a subacute left-sided course, not with injected skin flora and lung lesions.

B is wrong: Enterococcus faecalis endocarditis follows gastrointestinal or genitourinary manipulation, usually in older adults, and runs a subacute course.

D is wrong: Staphylococcus epidermidis infects prosthetic material as a biofilm, typically early after valve surgery, and this patient has a native valve.

A competent response contains

· Injection drug use, a short febrile course, and a holosystolic murmur at the left lower sternal border that increases with inspiration indicate tricuspid regurgitation from right-sided endocarditis.

· Cavitating peripheral lung nodules are septic pulmonary emboli from a right-sided vegetation.

· Staphylococcus aureus is the expected organism in injection drug use and the most common cause of endocarditis overall.

Distinctions that matter

· Right-sided vegetations embolize to the lungs; left-sided vegetations embolize to the systemic circulation.

Common misconceptions

· Reading cavitating lung nodules as a primary pneumonia. They are the first capillary bed downstream of the tricuspid valve.

Where it is taught: Section C01, organism table and the worked example of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C02-BD1

Brain dump 1 · foundational retrieval: Build the complication table by mechanism

A competent response contains

· Valve destruction: acute regurgitation, heart failure, chordal rupture. Clue is a new murmur with pulmonary edema.

· Perivalvular abscess: AV block and fistula between chambers. Clue is an aortic or prosthetic valve with new PR prolongation.

· Left-sided emboli: stroke, brain abscess, splenic and renal infarcts or abscesses, ischemic limbs or digits, Janeway lesions, splinter hemorrhages. Clue is a neurologic deficit or flank or left upper quadrant pain.

· Right-sided emboli: septic pulmonary emboli and lung abscess. Clue is injection drug use with cavitating nodules.

· Immune complexes: glomerulonephritis, Osler nodes, Roth spots. Clue is hematuria with red cell casts and low complement.

· Arterial wall infection: mycotic aneurysm. Clue is sudden headache or hemorrhage weeks into the illness.

· Ring abscess is most common with the aortic valve and with prosthetic valves, and the aortic annulus lies next to the AV node and the His bundle.

· The immune complex lesion is type III hypersensitivity: complexes deposit in glomeruli, activate complement, and cause a proliferative glomerulonephritis with hematuria and red cell casts and low complement levels; rheumatoid factor may be positive.

· Heart failure is the leading cause of death. Bacterial enzymes and neutrophil products perforate leaflets and tear chordae, and sudden severe regurgitation gives the ventricle no time to dilate, so pressure is transmitted back to the lungs and pulmonary edema follows.

Distinctions that matter

· Osler nodes are painful and attributed to immune complex vasculitis; Janeway lesions are painless and attributed to septic microemboli.

· A new murmur points to destroyed valve tissue; a new PR prolongation points to infection spreading into the annulus.

Common misconceptions

· Memorizing the peripheral signs as a list. Each one belongs to a mechanism, and the mechanism is what a question tests.

· Treating splinter hemorrhages as specific. They are linear and nonspecific and also follow trauma.

Where it is taught: Section C02, all four mechanism paragraphs and the mechanism table of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C02-BD2

Brain dump 2 · integration and application: Sort four new findings back to their mechanisms

A competent response contains

· Patient 1: left-sided emboli from a mitral or aortic vegetation. Fragments enter the systemic circulation and lodge in the first capillary bed downstream, giving splenic infarction and stroke.

· Patient 2: right-sided emboli from a tricuspid vegetation. Fragments reach the lungs as septic pulmonary emboli, producing multiple peripheral nodules that may cavitate, with cough, pleuritic pain, and hemoptysis.

· Patient 3: perivalvular (ring) abscess extending from the aortic valve. The aortic annulus lies next to the AV node and the His bundle, so a new and lengthening PR interval or a higher degree of AV block in a patient with endocarditis suggests an abscess.

· Patient 4: immune complexes. Continuous antigen release drives antibody production, complexes deposit in glomeruli and activate complement, giving hematuria with red cell casts and low complement, and the painful finger pad nodules are Osler nodes.

· Sudden severe headache in week 6 suggests rupture of a mycotic aneurysm with intracranial hemorrhage, typically in a branch of the middle cerebral artery.

· A septic embolus lodges in the vasa vasorum or lumen of an artery, infection weakens the wall, and the weakened wall dilates and can rupture.

· Sudden severe regurgitation meets a ventricle of normal size and compliance, so the regurgitant volume raises left atrial and pulmonary venous pressure sharply and pulmonary edema follows. Over years the ventricle dilates and accommodates the same volume at a lower filling pressure.

· Large, mobile vegetations embolize most often.

Distinctions that matter

· The side of the heart decides where emboli land, because emboli stop in the first capillary bed downstream.

· AV block in endocarditis is a structural complication of the annulus, separate from the valve lesion itself.

Common misconceptions

· Explaining hematuria in endocarditis as embolic every time. Red cell casts with low complement point to immune complex glomerulonephritis.

· Expecting chronic and acute regurgitation of the same severity to look alike. Time to remodel is what separates them.

Where it is taught: Section C02, paragraphs on local destruction, embolism, and immune complexes, with the mechanism table of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C02-CA01

Clinical application: Six weeks of fever, blood in the urine, and a changing murmur

A competent response contains

· Recognize: subacute infective endocarditis of the mitral valve, expected organism a viridans streptococcus from the dental extraction. The findings named are Osler nodes (painful finger pad nodules), Janeway lesions (painless macules on the palms), Roth spots (retinal hemorrhages with pale centers), and immune complex glomerulonephritis.

· Explain: the extraction released mouth flora into the blood; her long-standing regurgitant jet had already injured the atrial surface of the mitral valve and laid down a sterile platelet and fibrin deposit; viridans

streptococci adhere to that deposit through dextran and other adhesins, fibrin covers them, neutrophils cannot penetrate, and the vegetation grows and sheds organisms continuously.

· Differentiate: Janeway lesions are usually attributed to septic microemboli; Osler nodes, Roth spots, and the glomerulonephritis are attributed to immune complexes. Embolic lesions are occlusive and painless at the skin, while immune complex lesions are inflammatory and tender, and the low complement with red cell casts marks complex deposition rather than occlusion.

· Investigate: dysmorphic erythrocytes and erythrocyte casts with no bacteria mean the bleeding is glomerular rather than from the urinary tract, and low C3 and C4 mean complement has been consumed. This is a hypocomplementemic proliferative glomerulonephritis from type III (immune complex) hypersensitivity, with complexes deposited in the glomerular capillaries.

· Decide: a new and lengthening PR interval or a higher degree of AV block would indicate extension into the aortic annulus as a perivalvular abscess. A new loud murmur with sudden dyspnea and pulmonary edema would indicate that a leaflet has perforated or a chord has ruptured.

· Justify: the aortic annulus lies next to the AV node and the His bundle, so an abscess there interrupts conduction and lengthens the PR interval. A ventricle facing sudden severe regurgitation has had no time to dilate, so the extra volume raises left atrial and pulmonary venous pressure abruptly and fluid moves into the alveoli.

Distinctions that matter

· Glomerular bleeding gives dysmorphic cells and red cell casts; bleeding from the urinary tract does not.

· A changing murmur reports the valve, while a changing PR interval reports the annulus.

Common misconceptions

· Reading the renal failure here as embolic infarction. Low complement with red cell casts points to immune complex glomerulonephritis.

· Treating a 6-week course as evidence against endocarditis. A low-virulence organism on a damaged valve smolders for weeks.

Where it is taught: Section C02, with vegetation formation and organism prediction in Section C01 of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C03-BD1

Brain dump 1 · foundational retrieval: Trypanosoma cruzi from the bug to the tissue

A competent response contains

· Trypanosoma cruzi is a flagellated protozoan found in Mexico, Central America, and South America.

· The vector is the triatomine (reduviid, or kissing) bug, which lives in cracks in mud and thatch houses and feeds at night, often on the face.

· The bug defecates while or after feeding, and infective trypomastigotes in the feces are rubbed into the bite wound or onto the conjunctiva or other mucosa, so the feces and not the bite itself carry the parasite in.

· Other routes: blood transfusion, organ transplantation, mother to fetus, and food or drink contaminated with bug feces.

· Trypomastigotes invade local cells, transform into intracellular amastigotes, multiply, and burst out as trypomastigotes that spread through the blood.

· Amastigotes are the intracellular nonflagellated replicating form; trypomastigotes are the form that travels in the blood.

· Acute entry findings: a chagoma, an indurated skin lesion at a skin entry site, and Romana sign, painless unilateral swelling of the eyelids when entry is through the conjunctiva.

· The acute phase lasts weeks to about two months and most infections are mild or unnoticed; trypomastigotes are numerous enough to be seen on a blood smear.

· Acute myocarditis or meningoencephalitis occurs in a small minority, mainly young children and immunocompromised people.

· In nonendemic countries most cases are in immigrants from endemic regions.

Distinctions that matter

· Amastigote and trypomastigote are different stages with different locations: inside cells and multiplying, against in the blood and spreading.

· A chagoma marks a skin entry site and Romana sign marks a conjunctival entry site.

Common misconceptions

· Saying the bite inoculates the parasite. The parasite is in the bug's feces and is rubbed into the wound or the eye.

· Expecting severe acute illness. Most acute infections are mild or unnoticed, and severe acute disease is limited to young children and immunocompromised people.

Where it is taught: Section C03, first two paragraphs and the flow diagram of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C03-BD2

Brain dump 2 · integration and application: Explain each sign of chronic Chagas disease

A competent response contains

· Enlarged heart with low ejection fraction: decades of inflammation with patchy myocardial fibrosis and myocyte loss give a dilated cardiomyopathy.

· Right bundle branch block with left anterior fascicular block, and AV block: fibrosis has destroyed conduction tissue.

· Thin bulging apex: thinning of the left ventricular apex produces an apical aneurysm, in which mural thrombus forms and embolizes.

· Dysphagia: loss of myenteric (Auerbach) plexus neurons in the esophagus abolishes coordinated peristalsis and sphincter relaxation, giving an achalasia-like megaesophagus.

· Severe constipation: the same neuron loss in the colon gives megacolon.

· Parasites persist in cardiac muscle and enteric neurons; tissue sections show lymphocytic infiltrates and fibrosis with few parasites, and an autoimmune response contributes to the damage.

· Sudden death can be the first sign because scarred myocardium supports reentrant ventricular tachycardia.

· Parasitemia is now too low for a blood smear, so chronic disease is identified by antibodies against T. cruzi (serology).

· About 20 to 30 percent of infected people develop chronic disease, years to decades after a lifelong indeterminate phase in which the immune response controls but does not clear the parasite.

Distinctions that matter

· Acute disease is found by seeing the parasite; chronic disease is found by detecting antibody, because parasite numbers have fallen while tissue damage has grown.

· Conduction block and dilated cardiomyopathy come from the same fibrosis, but the block reflects where the fibrosis sits rather than how much muscle is lost.

Common misconceptions

· Expecting abundant parasites in chronic heart tissue. The sections show lymphocytes and fibrosis with few organisms.

· Treating the megaesophagus as a muscle problem. The neurons of the myenteric plexus are what has been lost.

Where it is taught: Section C03, third paragraph, the comparison table, and the final paragraph of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C03-CP01

Comparison table: Acute against chronic Chagas disease

A competent response contains

· Timing and share: acute is weeks after infection and affects everyone infected; chronic appears years to decades later in about 20 to 30 percent.

· Parasites: acute has abundant trypomastigotes in the blood; chronic has few, persisting in heart and gut tissue.

· Findings outside the heart: acute gives fever, chagoma, Romana sign, and lymphadenopathy, and is often without symptoms; chronic gives megaesophagus with dysphagia and megacolon with constipation.

· Heart: acute myocarditis is rare; chronic gives fibrosis, conduction block, apical aneurysm with thrombi, arrhythmia, and dilated cardiomyopathy.

· Detection: blood smear in the acute phase, serology in the chronic phase.

· Microscopy: acute tissue contains amastigotes in infected cells with inflammation; chronic tissue shows lymphocytic infiltrates and fibrosis with few parasites.

· Most infected people never progress beyond the indeterminate phase.

Distinctions that matter

· The two phases differ in parasite burden and in tissue damage in opposite directions, which is why the test changes with the phase.

· The indeterminate phase is a third state, not a mild form of either phase.

Common misconceptions

· Assuming chronic disease follows every infection. It develops in about 20 to 30 percent.

Where it is taught: Section C03, acute against chronic table of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C04-BD1

Brain dump 1 · foundational retrieval: Build the three stages of Lyme disease

A competent response contains

· Borrelia burgdorferi is a spirochete transmitted by Ixodes ticks: Ixodes scapularis, the black-legged or deer tick, in the eastern United States, and Ixodes pacificus on the Pacific coast.

· White-footed mice are the main reservoir and deer sustain the adult tick population.

· Risk is concentrated in the Northeast, the mid-Atlantic states, and the upper Midwest (Wisconsin and Minnesota), with a smaller Pacific coast focus.

· Most infections are acquired in late spring and summer from nymphs, which are tiny and easily missed.

· The spirochete sits in the tick midgut and must migrate to the salivary glands after feeding begins, so transmission usually requires attachment for more than about 36 hours.

· Early localized, days to about 1 month: erythema migrans with fever, fatigue, headache, and myalgia, from the spirochete spreading through the skin around the bite.

· Early disseminated, weeks to months: multiple erythema migrans lesions, facial nerve palsy (unilateral or bilateral), lymphocytic meningitis, radiculoneuritis, and carditis with AV block, from bloodstream spread to skin, nerves, meninges, and heart.

· Late, months to years: recurrent arthritis of large joints, especially the knee, and chronic neuropathy or encephalopathy, from persistent antigen and immune inflammation in synovium and nerve.

· Erythema migrans is an expanding red patch at least about 5 cm across that may clear centrally into a target shape.

· The same tick can transmit Babesia and Anaplasma, so coinfection is possible.

· Not every patient passes through every stage, and many do not recall the tick bite.

Distinctions that matter

· The stages follow where the spirochete is: in the skin around the bite, then in the blood reaching nerves, meninges, and heart, then persisting in synovium and nerve.

· Nymphs rather than adults transmit most infections, which is why the bite is so often unnoticed.

Common misconceptions

· Expecting a recalled tick bite. Nymphs are tiny and most patients do not remember one.

· Treating the attachment time as arbitrary. It reflects the migration of the spirochete from the tick midgut to the salivary glands.

Where it is taught: Section C04, first two paragraphs and the stages table of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C04-BD2

Brain dump 2 · integration and application: Lyme carditis and why the testing changes with the stage

A competent response contains

· This is early disseminated Lyme disease, which is when carditis occurs, in a small percentage of untreated patients.

· Spirochetes and a lymphocytic infiltrate inflame the myocardium and the conduction system, most often at the AV node.

· AV block of any degree can occur, from a prolonged PR interval to complete heart block, and the degree can change within hours because the tissue is inflamed rather than destroyed.

· The block usually resolves, because the node is inflamed rather than scarred. Myopericarditis can occur but heart failure is uncommon.

· Erythema migrans appears before antibodies are reliably detectable, so early localized disease is recognized clinically in an exposed patient.

· Disseminated and late disease are confirmed by serology, usually a screening immunoassay followed by a confirmatory test such as an immunoblot.

· Bilateral facial palsy is unusual and helps point to Lyme disease rather than idiopathic (Bell) palsy.

· Chagas disease also blocks conduction, but through fibrosis built over years with a dilated cardiomyopathy, apical aneurysm, and megaesophagus or megacolon, in someone exposed in Latin America.

· A perivalvular abscess from endocarditis also blocks conduction, but the patient has a fever with an infected aortic or prosthetic valve, a murmur, and a PR interval that lengthens as the abscess extends.

Distinctions that matter

· Inflammation reverses and scar does not, which is why Lyme block resolves while Chagas block persists.

· Testing follows the antibody response rather than the symptom: too early for serology in the skin stage, reliable once the infection has disseminated.

Common misconceptions

· Ordering serology to confirm erythema migrans. Antibodies are not yet reliably detectable at that stage.

· Assuming complete heart block means permanent conduction damage. In Lyme carditis the node is inflamed and the block usually resolves.

Where it is taught: Section C04, carditis and diagnosis paragraphs, with Section C02 on ring abscess and Section C03 on Chagas disease of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C04-Q02

Board-style question: Heart block after a summer rash

Correct answer: A. Lymphocytic inflammation of the atrioventricular node around a disseminated spirochete

Why it is right: Borrelia burgdorferi disseminates from the skin through the blood to the heart, and spirochetes with a lymphocytic infiltrate inflame the myocardium and the conduction system at the AV node, so AV block of any degree appears and can change within hours; the facial palsy and the preceding erythema migrans place her in the same stage.

B is wrong: Fibrosis of the conduction system from decades of Trypanosoma cruzi persistence requires exposure in Latin America and years of infection, and it comes with a dilated cardiomyopathy, apical aneurysm, and megaesophagus or megacolon.

C is wrong: A ring abscess from endocarditis occurs in a febrile patient with an infected aortic or prosthetic valve and a murmur, and the PR interval lengthens progressively rather than appearing after a summer rash.

D is wrong: Immune complex deposition in endocarditis injures glomeruli and small vessels, producing hematuria with red cell casts, Osler nodes, and Roth spots, not block of the conduction system.

A competent response contains

· A 9 cm expanding patch with central clearing in August in rural Wisconsin is erythema migrans, which marks Lyme disease in an endemic area.

· P waves at 82/min unrelated to QRS complexes at 40/min are complete AV block, and the narrow QRS places the escape pacemaker at the junction.

· A facial palsy that includes the forehead is a peripheral seventh nerve palsy, another feature of early disseminated disease.

· Carditis in early dissemination comes from spirochetes and a lymphocytic infiltrate inflaming the conduction system, most often at the AV node.

Distinctions that matter

· Rash, facial palsy, and heart block together place the illness in the early disseminated stage.

Common misconceptions

· Reading complete heart block in a young patient as degenerative conduction disease. The exposure history and the rash point to a treatable infection of the node.

Where it is taught: Section C04 of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C05-BD1

Brain dump 1 · foundational retrieval: Rickettsia rickettsii from the tick to the vessel wall

A competent response contains

· Rickettsia rickettsii is a small obligate intracellular gram-negative bacterium that stains poorly with Gram stain, and it must be inside a host cell to replicate.

· Vectors are Dermacentor ticks: the American dog tick in the east and the Rocky Mountain wood tick in the west, with the brown dog tick in parts of Arizona and Mexico.

· Most United States cases occur in the southeastern and south-central states, such as North Carolina, Tennessee, Arkansas, Missouri, and Oklahoma, in spring and summer.

· Symptoms begin about 2 to 14 days after the bite.

· The target cell is the vascular endothelial cell. Rickettsiae attach through outer membrane proteins, are taken up, escape the phagosome into the cytoplasm, multiply there, and polymerize host actin into tails that push them into neighboring endothelial cells, so infection spreads cell to cell along the vessel.

· The lesion is a disseminated small vessel vasculitis, reaching skin, brain, lungs, kidneys, heart, and gut.

· Rash: appears about day 2 to 5 as small blanching pink macules on the wrists and ankles, spreads centripetally to the arms, legs, and trunk, and becomes petechial as red cells leak from damaged capillaries.

· Involvement of the palms and soles is characteristic but often late.

Distinctions that matter

· Spread cell to cell by actin tails keeps the organism inside the endothelium, which is why the disease is a vasculitis rather than a bacteremic illness with abscesses.

· The name points to the Rocky Mountains while the cases are concentrated in the southeast and south-central states.

Common misconceptions

· Expecting the rash first. Fever and headache come first and the rash appears about day 2 to 5, and some patients never develop one.

· Describing the rash as spreading outward from the trunk. It starts at the wrists and ankles and spreads centripetally.

Where it is taught: Section C05, first two paragraphs, the flow diagram, and the rash paragraph of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C05-BD2

Brain dump 2 · integration and application: Derive every laboratory finding from the leaking vessel

A competent response contains

· Thrombocytopenia: platelets adhere to damaged endothelium and exposed subendothelium and are consumed there. The marrow is producing platelets normally; they are being used up at the vessel wall.

· Hypoalbuminemia with edema: fluid and albumin leak through damaged capillaries into the interstitium.

· Hyponatremia: the fall in effective circulating volume stimulates ADH release, so water is retained and the serum sodium falls.

· Petechial rash: red cells escape from damaged capillaries, so the blanching macules become nonblanching petechiae.

· Noncardiogenic pulmonary edema: the same leak occurs in pulmonary capillaries, so fluid enters the alveoli with normal cardiac filling pressures.

· Confusion and seizures: vasculitis in cerebral vessels with edema; coma can follow in severe disease.

· Shock and acute kidney injury: hypovolemia from plasma loss, with myocardial involvement, lowers perfusion pressure, and the kidney is injured by both hypoperfusion and its own vasculitis. Liver enzymes may rise, and disseminated intravascular coagulation and gangrene of digits can occur but are less common.

· The classic triad is fever, headache, and rash with a history of a tick bite or exposure; all three are present at first in only a minority.

· The diagnosis is made on clinical grounds because the course can be rapidly fatal and antibodies are often absent in the first week.

· Other rashes on the palms and soles: secondary syphilis (copper-colored macules or papules, generalized lymphadenopathy, condylomata lata), hand, foot, and mouth disease from coxsackie A (vesicles with oral ulcers, usually in young children), and infective endocarditis (Janeway lesions with fever and a murmur).

Distinctions that matter

· Low platelets here are consumptive, not a production problem, which is why the marrow is normal.

· Myalgia, nausea, vomiting, and abdominal pain are common and can mislead toward a gastrointestinal illness.

Common misconceptions

· Explaining the hyponatremia as sodium loss. Water is retained because hypovolemia drives ADH release.

· Waiting for serology before treating the illness as spotted fever. Antibodies are often absent in the first week and the course can be rapidly fatal.

Where it is taught: Section C05, laboratory paragraph, triad paragraph, and the palms and soles table of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C05-PR01

Prediction exercise: Predict what happens as the vasculitis spreads

A competent response contains

· Permeability rises and interstitial fluid increases, because infected endothelial cells and the inflammation around them no longer hold the barrier.

· Serum albumin falls and effective circulating volume falls, because fluid and protein escape through the damaged wall.

· ADH release rises and the serum sodium falls, because the drop in effective circulating volume is a stimulus for ADH.

· The platelet count falls, and the vessel wall accounts for it: platelets adhere to damaged endothelium and are consumed, with the marrow still producing.

· The lesions become petechial and nonblanching as red cells escape, appearing first on the wrists and ankles.

· Oxygenation falls as fluid enters the alveoli, which is noncardiogenic pulmonary edema from leaking pulmonary capillaries.

· Blood pressure falls and urine output falls, because continued plasma loss causes hypovolemia and shock, with acute kidney injury from hypoperfusion and renal vasculitis.

Distinctions that matter

· Every item traces to one event at the vessel wall, so the prediction is the same reasoning applied to different organs.

· Pulmonary edema here is permeability edema with normal cardiac filling pressures, unlike cardiogenic edema.

Common misconceptions

· Predicting a rising hematocrit and a rising albumin together. Albumin is lost with the plasma, so it falls.

· Predicting marrow failure from the low platelet count. Platelets are consumed at the injured endothelium.

Where it is taught: Section C05, flow diagram and laboratory paragraph, with the stop and predict box of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C05-Q03

Board-style question: Low sodium and low platelets in a spotted fever

Correct answer: D. Infection of vascular endothelium causing plasma leak and platelet consumption at the vessel wall

Why it is right: Rickettsiae replicate inside endothelial cells and push into neighboring cells using host actin, so small vessels throughout the body become inflamed and leaky: plasma and albumin escape into the interstitium giving edema and hypoalbuminemia, the fall in effective circulating volume releases ADH

and lowers the serum sodium, red cells escape to give petechiae, and platelets adhere to the damaged wall and are consumed.

A is wrong: Immune complex deposition with complement activation is the mechanism of the glomerulonephritis, Osler nodes, and Roth spots of endocarditis, and would come with a murmur and low complement rather than tick exposure and a centripetal petechial rash.

B is wrong: Rickettsia rickettsii replicates in endothelium rather than in platelets or megakaryocytes, and the marrow continues to produce platelets while they are consumed peripherally.

C is wrong: Antibody-dependent enhancement explains severe dengue in a second infection with a different serotype, and this patient has a tick exposure with a rash that began at the wrists and ankles.

A competent response contains

· Tick exposure in North Carolina in May with fever, severe headache, and petechiae that began at the wrists and ankles identifies Rocky Mountain spotted fever.

· Rickettsia rickettsii infects vascular endothelium and spreads cell to cell, producing a disseminated small vessel vasculitis.

· Edema with hypoalbuminemia comes from plasma leak, hyponatremia from ADH released in response to the fall in effective circulating volume, and thrombocytopenia from platelet consumption at the damaged wall.

Distinctions that matter

· One lesion at the endothelium explains all three laboratory abnormalities, which is what the question is testing.

Common misconceptions

· Looking for a separate cause for each abnormal value. They share a single vascular mechanism.

Where it is taught: Section C05 of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C06-BD1

Brain dump 1 · foundational retrieval: Yellow fever from the mosquito to the liver

A competent response contains

· Yellow fever and dengue are flaviviruses, enveloped positive-sense single-stranded RNA viruses, carried mainly by Aedes aegypti, a daytime-biting mosquito that breeds in standing water near houses.

· Yellow fever occurs in tropical Africa and South America. A jungle (sylvatic) cycle between monkeys and forest mosquitoes maintains the virus, and Aedes aegypti carries it between people in towns.

· After 3 to 6 days the illness begins with fever, headache, myalgia, and back pain, and most patients recover.

· About 15 percent improve briefly and then enter a toxic phase.

· The virus damages hepatocytes especially in the middle of the lobule, zone 2, which is midzonal necrosis. This is unusual because most toxic and ischemic injury is centrilobular.

· Dying hepatocytes undergo apoptosis and appear as Councilman bodies, shrunken eosinophilic cells, with microvesicular fat and relatively little inflammation.

· Liver failure causes jaundice, which gives the disease its name, and loss of clotting factor synthesis combines with platelet consumption to cause bleeding.

· Blood swallowed or vomited after gastric bleeding is darkened by acid, which produces black vomit.

· Kidney failure and shock follow, and a large share of patients in the toxic phase die.

· Laboratory clues are very high aminotransferases and bilirubin with a long prothrombin time.

Distinctions that matter

· Midzonal necrosis separates yellow fever from the centrilobular pattern of most toxic and ischemic liver injury.

· Councilman bodies are apoptotic hepatocytes, which is why inflammation is relatively sparse.

Common misconceptions

· Expecting a heavy inflammatory infiltrate in the yellow fever liver. The injury is apoptotic with relatively little inflammation.

· Explaining the bleeding by liver failure alone. Loss of clotting factor synthesis and platelet consumption act together.

Where it is taught: Section C06, opening paragraph and the yellow fever paragraph, with the comparison table of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C06-BD2

Brain dump 2 · integration and application: Why the second dengue infection is the dangerous one

A competent response contains

· Dengue has four serotypes, DENV-1 to DENV-4.

· Antibodies against the earlier serotype bind the new serotype but do not neutralize it.

· The Fc portion of the bound antibody attaches to Fc gamma receptors on monocytes and macrophages, so more virus enters these cells than would enter without antibody; viral load and cytokine release rise.

· This is antibody-dependent enhancement, and infants carrying maternal antibody are the other group at risk through the same route.

· Severe dengue usually appears around defervescence, about days 3 to 7, as the fever falls.

· At the capillary wall, endothelial activation by viral proteins and cytokines makes capillaries leak plasma.

· A hematocrit rising from 40% to 48% means plasma has left the vessels while red cells stayed behind, which is hemoconcentration.

· It warns of shock before the blood pressure falls because intravascular volume is already being lost while compensation still maintains pressure.

· As the leak continues, fluid collects as pleural effusions and ascites, and hypovolemic shock can follow, while platelets fall and bleeding may occur.

· Classic dengue causes sudden high fever, often 39 to 40 °C (102.2 to 104 °F), headache with pain behind the eyes, rash, and severe muscle, joint, and bone pain (breakbone fever), with leukopenia and thrombocytopenia.

Distinctions that matter

· The rising hematocrit measures plasma leaving the vessels, while the falling platelet count measures consumption, so the two findings report different processes.

· Severe dengue begins as the fever settles rather than at the height of the fever.

Common misconceptions

· Reading a rising hematocrit as improvement. It is a sign of plasma loss and warns of shock.

· Assuming the first infection is the dangerous one. Pre-existing non-neutralizing antibody is what makes the second infection with a different serotype worse.

Where it is taught: Section C06, dengue paragraph and the worked example on the second infection of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C06-CP01

Comparison table: Yellow fever against dengue

A competent response contains

· Where: yellow fever in tropical Africa and South America; dengue throughout the tropics and subtropics worldwide.

· Main target: hepatocytes for yellow fever; monocytes, macrophages, and dendritic cells for dengue, with endothelium affected.

· Hallmark pathology: midzonal necrosis with Councilman bodies for yellow fever; capillary plasma leak for dengue.

· Key findings: jaundice, black vomit, bleeding, and renal failure for yellow fever; severe myalgia and arthralgia, retro-orbital pain, and rash for dengue.

· Laboratory clue: very high aminotransferases and bilirubin with a long prothrombin time for yellow fever; low platelets, low white cells, and a rising hematocrit for dengue.

· Severe form linked to: the toxic phase after a brief remission for yellow fever; a second infection with a different serotype for dengue.

· Both are Aedes-borne flaviviruses with thrombocytopenia and bleeding, and the liver is what separates them.

Distinctions that matter

· Both viruses share a family and a mosquito, so the organ damaged is what identifies the disease.

· Bleeding occurs in both, but from failed clotting factor synthesis in yellow fever and from consumption with endothelial injury in dengue.

Common misconceptions

· Grouping both diseases as the same hemorrhagic illness. The hepatitis of yellow fever and the plasma leak of dengue produce different pictures.

Where it is taught: Section C06, yellow fever against dengue table of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C07-BD1

Brain dump 1 · foundational retrieval: The three shared effects of hemorrhagic fever viruses

A competent response contains

· Viral hemorrhagic fever is a syndrome rather than one disease, caused by enveloped RNA viruses from several families.

· Flaviviruses (yellow fever, dengue), positive-sense single-stranded RNA; filoviruses (Ebola, Marburg), negative-sense single-stranded RNA and filamentous; arenaviruses (Lassa), ambisense segmented single-stranded RNA; bunyaviruses (hantaviruses, Crimean-Congo hemorrhagic fever), negative-sense segmented single-stranded RNA.

· First effect: the viruses infect monocytes, macrophages, and dendritic cells early and spread with them to lymph nodes, liver, and spleen. Many block type I interferon signaling, so replication outpaces the innate response, infected dendritic cells fail to start an effective adaptive response, and many lymphocytes die by bystander apoptosis.

· Second effect: infected macrophages release large amounts of TNF-alpha, IL-1, IL-6, and other mediators, a cytokine storm, and these, with direct infection of endothelium in some viruses, loosen endothelial junctions and increase permeability.

· Third effect: activated macrophages express tissue factor, which starts coagulation throughout the circulation; platelets and clotting factors are consumed faster than they are made, which is consumptive coagulopathy, also called disseminated intravascular coagulation.

· Hepatocyte injury lowers clotting factor production further.

· The patient therefore leaks plasma and bleeds at the same time.

· Despite the name, visible bleeding is often modest; most deaths come from loss of fluid out of the vessels, shock, and multiorgan failure.

Distinctions that matter

· Permeability and coagulopathy are separate effects of the same infected macrophage, which is why shock and bleeding appear together.

· Interferon blockade is about timing: it lets replication run ahead of the host response.

Common misconceptions

· Attributing the deaths to blood loss. Plasma leak with shock and multiorgan failure accounts for most of them.

· Treating all of these viruses as one family. They come from several families and share mechanisms rather than taxonomy.

Where it is taught: Section C07, first two paragraphs, the flow diagram, and the virus table of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C07-BD2

Brain dump 2 · integration and application: Work out who is at risk in a filovirus outbreak

A competent response contains

· Ebola and Marburg are filoviruses: enveloped, nonsegmented, negative-sense single-stranded RNA viruses with long filamentous virions. Fruit bats are the likely reservoir.

· Outbreaks begin when a person handles an infected animal or its tissues, or, for Marburg, enters bat-inhabited caves or mines. They occur in sub-Saharan Africa.

· Spread between people is by direct contact with blood, vomit, stool, or other body fluids of a symptomatic patient or a body after death, or with contaminated objects.

· Health care workers are at high risk through contact with the body fluids of symptomatic patients, and people preparing bodies for burial are at high risk through contact with the body after death.

· Patients become infectious when symptoms start, viral load rises as the illness worsens, and virus can persist in semen and other protected sites for months after recovery.

· Ordinary airborne spread does not occur.

· Incubation is about 2 to 21 days. Illness starts with sudden fever, weakness, myalgia, and headache, followed by profuse vomiting and diarrhea; a maculopapular rash, conjunctival injection, and bleeding from gums, venipuncture sites, or the gut may appear in the second week.

· The profuse vomiting and diarrhea account for most of the fluid and electrolyte loss.

· Laboratory findings: thrombocytopenia, leukopenia, high aminotransferases, and evidence of coagulopathy and kidney injury.

· Case fatality is about 50 percent on average for Ebola, ranging widely between outbreaks, and similarly high for Marburg.

Distinctions that matter

· Infectiousness begins with symptoms, which is why contact tracing focuses on symptomatic patients and on bodies after death.

· Persistence in semen and other protected sites is a separate route from the acute illness.

Common misconceptions

· Describing these viruses as airborne. Ordinary airborne spread does not occur; transmission is by direct contact with body fluids.

· Attributing the volume loss to bleeding. Vomiting and diarrhea account for most of it.

Where it is taught: Section C07, filovirus paragraphs and the virus table of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C07-RT01

Retrieval exercise: Rebuild the common pathway of the hemorrhagic fever viruses

A competent response contains

· Step 1: the virus infects macrophages, dendritic cells, and in some cases endothelium.

· Step 2: interferon signaling is blocked and replication spreads to liver, spleen, and lymph nodes.

· Step 3: cytokine storm with TNF-alpha, IL-1, and IL-6 released.

· Step 4: endothelial junctions loosen, so plasma leaks and hypovolemia develops.

· Step 5: macrophage tissue factor triggers consumptive coagulopathy.

· Step 6: low platelets and clotting factors with liver injury cause bleeding.

· Step 7: shock and multiorgan failure.

· Blocking interferon signaling buys the virus time, so replication outpaces the innate response and the adaptive response starts late and poorly.

· Tissue factor is expressed by activated macrophages, and the consumption of platelets and clotting factors is consumptive coagulopathy, also called disseminated intravascular coagulation.

· Volume is lost at step 4 through the leaking endothelium and again at step 7 as shock deepens; bleeding becomes likely at step 5 when factors and platelets are consumed and at step 6 when liver injury lowers factor synthesis.

· Shock and bleeding appear together because the same activated macrophage both releases the cytokines that loosen the junctions and expresses the tissue factor that consumes the clotting system.

Distinctions that matter

· Leak and coagulopathy run in parallel from one cell, rather than one being a late consequence of the other.

Common misconceptions

· Drawing bleeding as the first event. Plasma leak and consumption of the clotting system come first, and bleeding follows.

Where it is taught: Section C07, flow diagram and the paragraph on the three shared effects of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C08-BD1

Brain dump 1 · foundational retrieval: Build the course of viral myocarditis

A competent response contains

· Myocarditis is inflammation of the myocardium with injury to myocytes that is not caused by ischemia, so ischemia must be excluded as the cause.

· Viruses are the most common cause in developed countries. Coxsackie B virus is the classic cause: an enterovirus, a nonenveloped positive-sense RNA picornavirus spread by the fecal-oral and respiratory routes.

· Adenovirus, parvovirus B19, human herpesvirus 6, influenza, and SARS-CoV-2 are also important.

· The typical patient is a young adult, more often male, with cardiac symptoms one to a few weeks after a respiratory or gastrointestinal illness, because the virus first replicates in the pharynx or gut and then reaches the heart through the blood.

· Course: viral infection of pharynx or gut; viremia with coxsackie B entering myocytes through CAR; direct viral injury to myocytes; T-cell attack with a lymphocytic infiltrate and myocyte necrosis; most recover while some develop persistent inflammation; fibrosis and dilation give dilated cardiomyopathy.

· The receptor is the coxsackie-adenovirus receptor (CAR), a junctional protein on cardiac myocytes, used by coxsackie B viruses and adenoviruses.

· CD8 cytotoxic T cells and natural killer cells destroy infected myocytes, and cytokines depress contractility.

· In the minority with persistent inflammation, autoimmunity can develop against cardiac proteins such as myosin, and progressive myocyte loss with fibrosis leaves a dilated cardiomyopathy.

· Parvovirus B19 infects coronary microvascular endothelium rather than myocytes.

Distinctions that matter

· Direct viral injury dominates the first days; the immune response dominates the following weeks.

· Most patients recover fully, and the minority with persistent inflammation are the ones at risk of chronic dilated cardiomyopathy.

Common misconceptions

· Attributing all the myocyte loss to the virus. Cytotoxic T cells and natural killer cells do much of the killing.

· Expecting cardiac symptoms during the viral illness. They follow it by one to a few weeks.

Where it is taught: Section C08, first two paragraphs and the flow diagram of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C08-BD2

Brain dump 2 · integration and application: Separate myocarditis from an infarction and from the other infiltrates

A competent response contains

· Troponin is raised because necrotic myocytes release it, and the myocytes are dying from inflammation rather than ischemia, which is why the coronary arteries are normal.

· Inflamed tissue disturbs conduction and repolarization, so the ECG shows ST and T changes, AV or bundle branch block, or ventricular arrhythmias.

· Myocarditis is an important cause of sudden death in young athletes.

· When the pericardium is involved (myopericarditis), chest pain is pleuritic and the ECG may show diffuse ST elevation.

· Loss of contracting myocytes lowers ejection fraction and causes heart failure with dyspnea, fatigue, and edema.

· Endomyocardial biopsy shows an inflammatory infiltrate, mainly lymphocytes, with necrosis of adjacent myocytes in a pattern that is not ischemic.

· A single biopsy can miss the disease because the inflammation is patchy.

· Cardiac MRI often shows edema and late gadolinium enhancement in the subepicardial or midwall layers, unlike the subendocardial or transmural pattern of infarction.

· Many eosinophils suggest hypersensitivity or eosinophilic myocarditis, and multinucleated giant cells suggest the rapidly progressive giant cell myocarditis.

· Because chest pain, ST elevation, and raised troponin can mimic infarction, the clue is a young patient with a recent viral illness and normal coronary arteries.

Distinctions that matter

· Raised troponin reports myocyte death without saying what killed the cell, so the coronary anatomy and the MRI pattern are what separate inflammation from ischemia.

· Diffuse ST elevation with pleuritic pain points to pericardial involvement, while infarction gives changes in the leads of one territory.

Common misconceptions

· Taking a negative biopsy as proof there is no myocarditis. Inflammation is patchy and a single sample can miss it.

· Reading raised troponin with ST elevation as infarction in every patient. A young patient with a recent viral illness and normal coronary arteries fits myocarditis.

Where it is taught: Section C08, findings paragraph and the biopsy and imaging paragraph, with the stop and predict box of the CV-18-K Infectious Vascular Diseases notes.

NPLEX-P1-CARD-18-11-C08-Q04

Board-style question: Biopsy after a viral illness

Correct answer: B. A lymphocytic infiltrate with necrosis of adjacent myocytes

Why it is right: A gastrointestinal illness a week earlier fits coxsackie B, which reaches the heart through the blood and enters myocytes through the coxsackie-adenovirus receptor; the immune response then

dominates, with CD8 cytotoxic T cells and natural killer cells killing infected myocytes, so the biopsy shows a mainly lymphocytic infiltrate with necrosis of the myocytes beside it.

A is wrong: Multinucleated giant cells with extensive myocyte loss indicate giant cell myocarditis, which is rapidly progressive and not the expected finding after a self-limited viral illness.

C is wrong: Intracellular amastigotes with interstitial fibrosis indicate chronic Chagas disease, which requires exposure in Latin America and develops years to decades after infection.

D is wrong: An eosinophil-rich infiltrate with little myocyte necrosis suggests hypersensitivity or eosinophilic myocarditis rather than a viral cause.

A competent response contains

· A young man with pleuritic chest pain, diffuse ST elevation, raised troponin, global hypokinesis, and normal coronary arteries one week after a febrile diarrheal illness has viral myocarditis, most often from coxsackie B.

· Coxsackie B enters myocytes through the coxsackie-adenovirus receptor, and CD8 cytotoxic T cells and natural killer cells destroy the infected cells.

· Biopsy therefore shows a lymphocytic infiltrate with necrosis of adjacent myocytes in a pattern that is not ischemic.

Distinctions that matter

· The histologic pattern names the cause: lymphocytes for viral myocarditis, eosinophils for hypersensitivity, giant cells for giant cell myocarditis.

Common misconceptions

· Expecting an ischemic pattern of necrosis because the troponin is raised and the ST segments are elevated.

Where it is taught: Section C08 of the CV-18-K Infectious Vascular Diseases notes.